XAI3 — Partial Dependence Plots

Diego · Haoxiang | May 2026

Part 1 — 1D PDPs: Bike Rentals

A Random Forest (ranger, 300 trees, mtry=2) was trained on day.csv to predict daily bike rentals (cnt). Variables were denormalised: temperature ×47−8 (°C), humidity ×100 (%), wind speed ×67 (km/h). The model achieves $R^2 \approx 0.88$ (RMSE≈671 bikes).

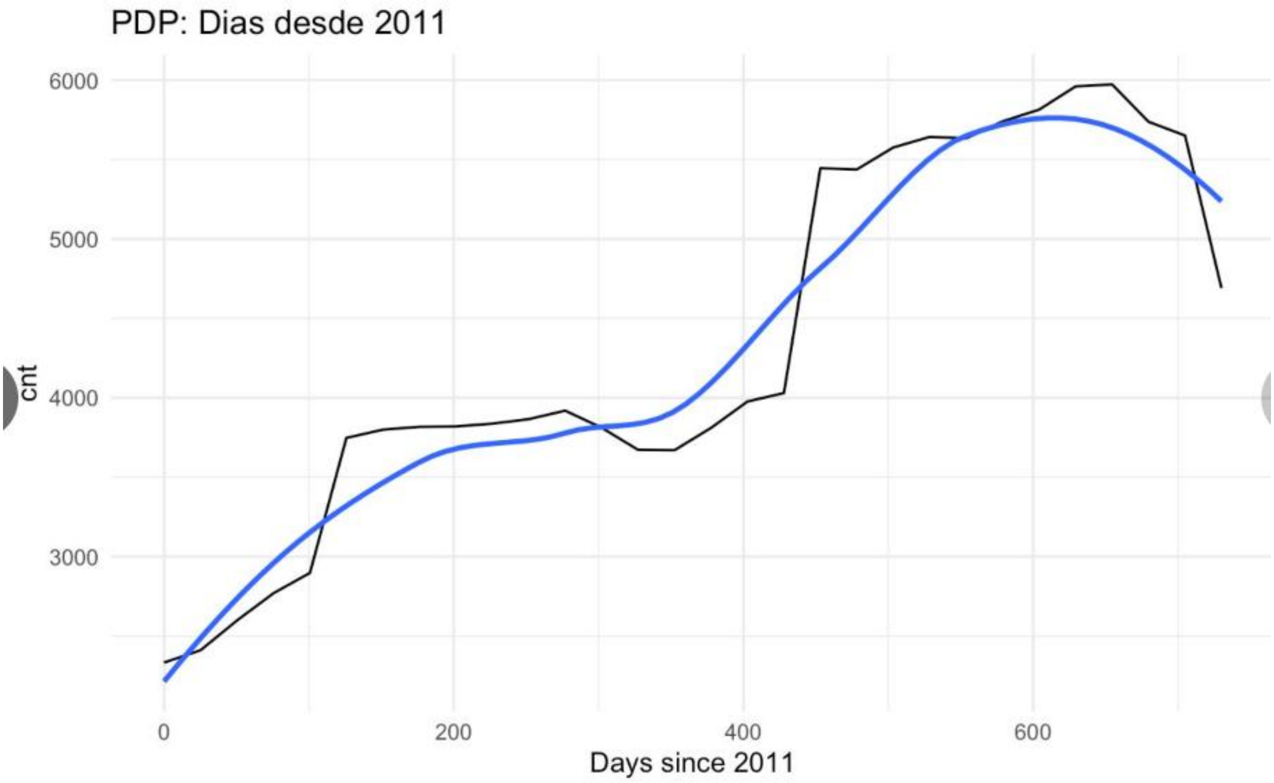
Metric	Value
Trees	300
mtry	2
R^2 (OOB)	≈ 0.88
RMSE	≈ 671 bikes
Target variable	cnt (total daily rentals)
Features used	days_since_2011, tempD, humD, windspeedD

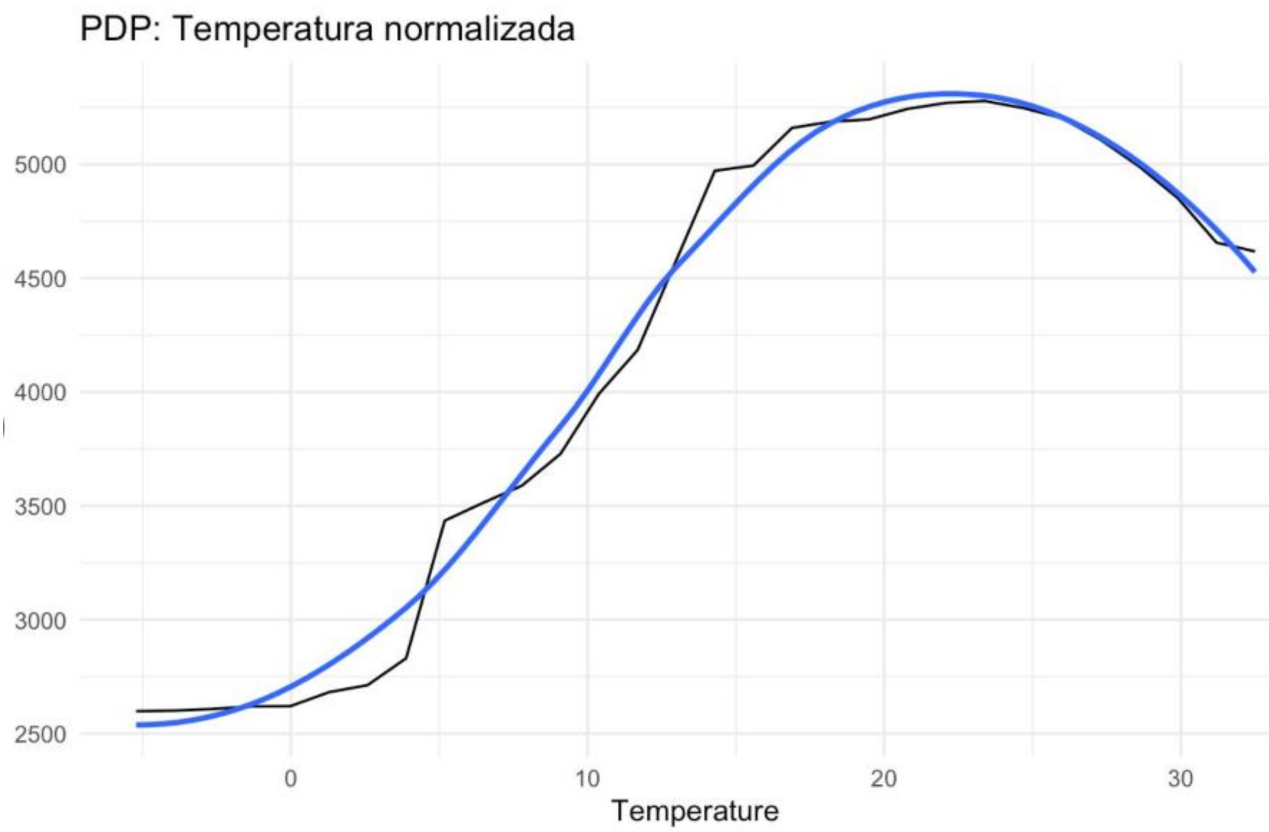
PDP Results

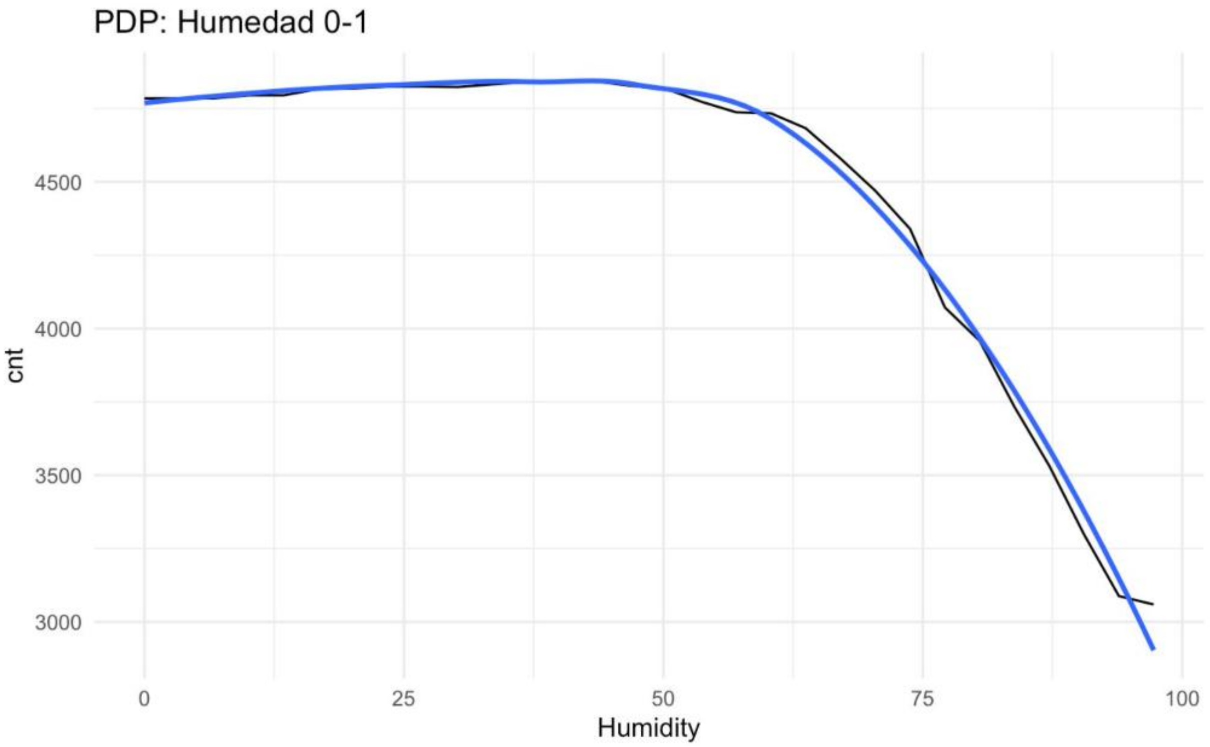
Feature	Effect	Key finding
days_since_2011	Strong positive trend	Rentals grow from ~2,000 to 5,000+ over two years
tempD (°C)	Inverted-U	Peak at 20–25 °C; drops at extreme cold or heat
humD (%)	Negative above 60%	Stable below 60% humidity, then falls sharply
windspeedD (km/h)	Monotonically negative	From ~4,750 (calm) down to ~4,000 at max wind

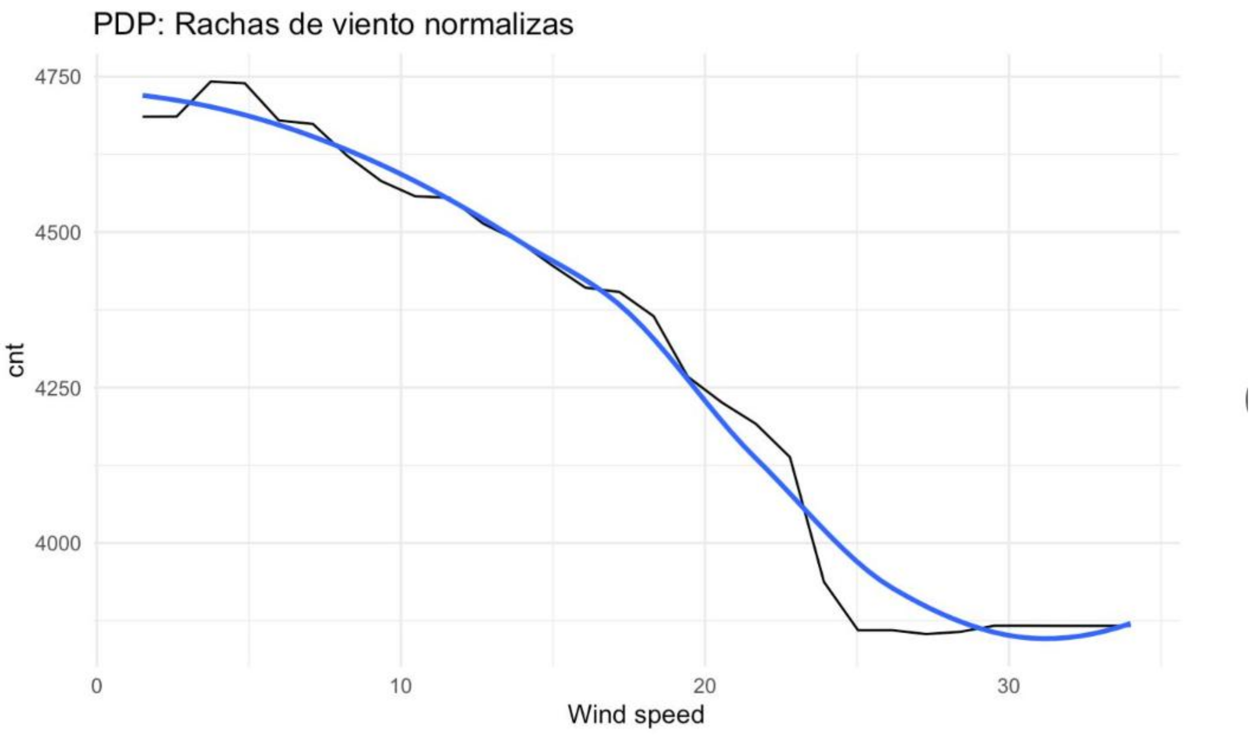
Plots

`geom_smooth()` using method = 'loess'



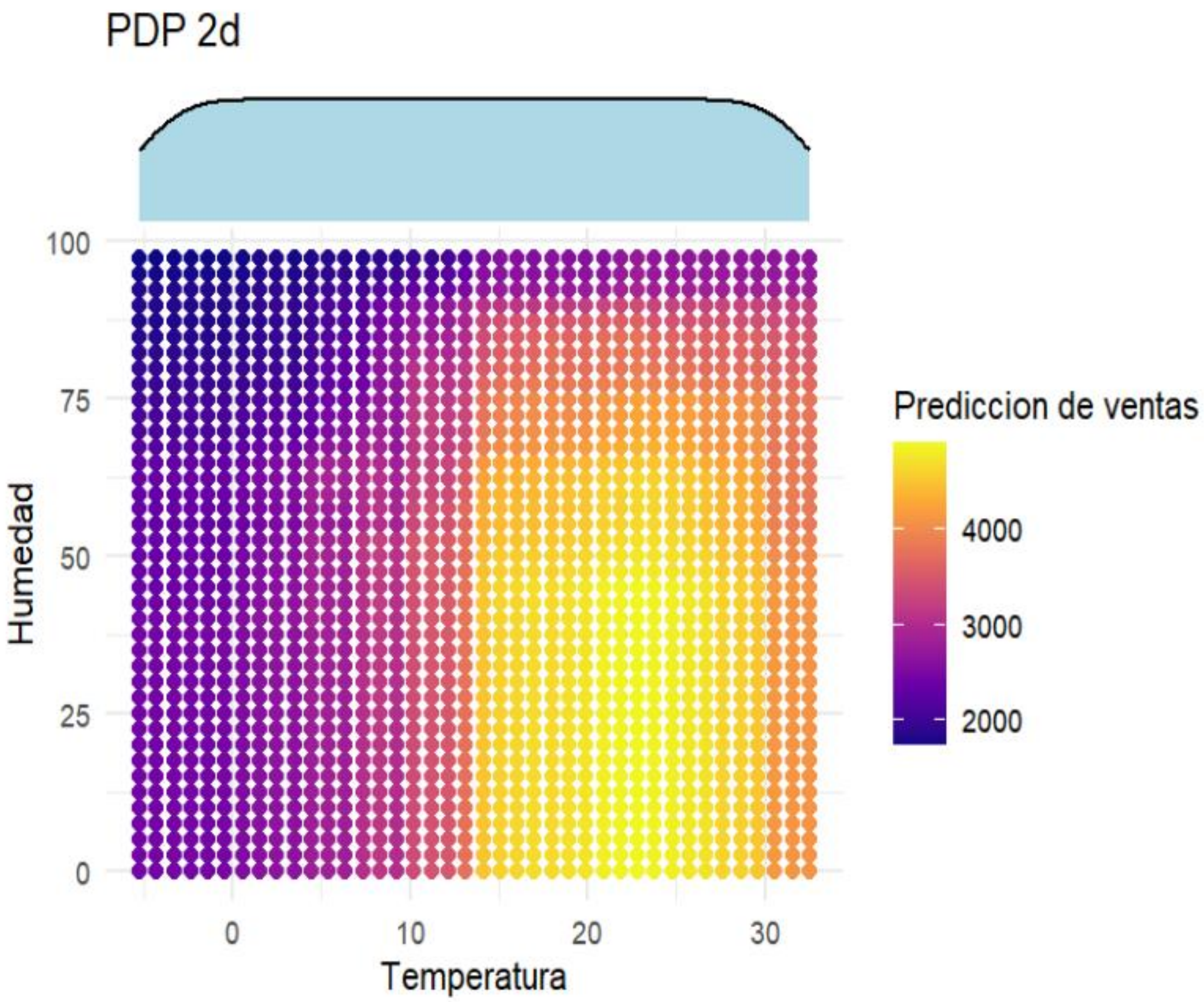






Part 2 — 2D PDP: Temperature × Humidity

A 2D PDP was generated on a 40×40 grid, fixing days_since_2011 and windspeedD at their medians. Marginal density distributions were overlaid with ggMarginal(). The highest predicted rentals occur where tempD > 20 °C and humD < 60%. Low temperature suppresses demand regardless of humidity level.



El gráfico de mosaico (geom_tile) muestra la predicción conjunta de cnt para cada combinación de temperatura (eje X) y humedad (eje Y). Las zonas más brillantes (amarillo-verde) corresponden a los valores más altos de rentas previstas, mientras que las oscuras (morado) indican pocas rentas.

Región óptima: Se localiza en el cuadrante de temperatura moderada-alta (≈0.5-0.7) y humedad baja (≈0.2-0.4). Allí las predicciones superan las 4500-5000 bicicletas. Es el clima ideal: cálido pero no bochornoso, seco.

Regiones con muy bajas predicciones (<2000):

Temperatura fría + humedad alta (esquina inferior izquierda, frío y húmedo).

Temperatura muy alta + humedad muy alta (esquina superior derecha, calor y bochorno).

También temperatura fría + humedad baja produce valores modestos (~2000), pero no tan extremos como con humedad alta.

Efecto de interacción: Para una misma temperatura, elevar la humedad siempre reduce las predicciones, pero el impacto es mayor cuando la temperatura ya es elevada. Por ejemplo, pasar de humedad 0.3 a 0.8 a temperatura 0.6 puede hacer caer la previsión de 5000 a 1500. En cambio, a temperatura baja (0.2) el mismo aumento de humedad solo baja de ~2000 a ~1000.

Las densidades marginales (histogramas/densidades en los bordes) indican dónde hay más datos reales. Observamos que la mayoría de los días tienen temperaturas entre 0.3 y 0.7, y humedades entre 0.4 y 0.7. La región óptima (cálida-seca) tiene menor densidad de datos, lo que sugiere que esas condiciones son menos frecuentes; sin embargo, cuando ocurren, generan una demanda muy alta. Por el contrario, las condiciones extremas (muy frío con alta humedad) son escasas, por lo que las predicciones en esas zonas deben tomarse con más precaución.

Conclusión final: El número de bicicletas alquiladas responde principalmente a un clima templado y seco, con un efecto claramente creciente a lo largo de los dos años. La humedad alta y el viento fuerte perjudican la demanda, y la temperatura tiene un punto óptimo intermedio. El modelo ha capturado estas relaciones de forma coherente y con gran poder predictivo.

Part 3 — PDPs: House Price Prediction

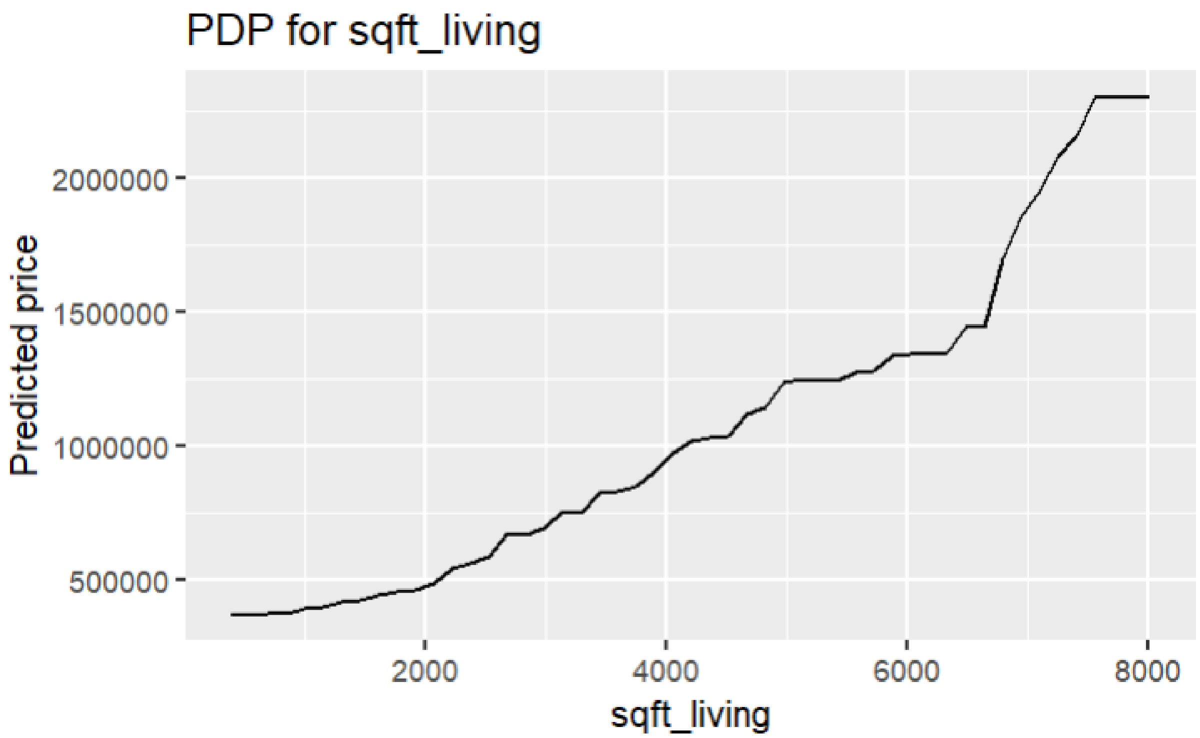
A Random Forest (randomForest, 100 trees) was trained on a subsample of 1,500 observations from kc_house_data.csv. Features: bedrooms, bathrooms, sqft_living, sqft_lot, floors, yr_built. Target: price (USD).

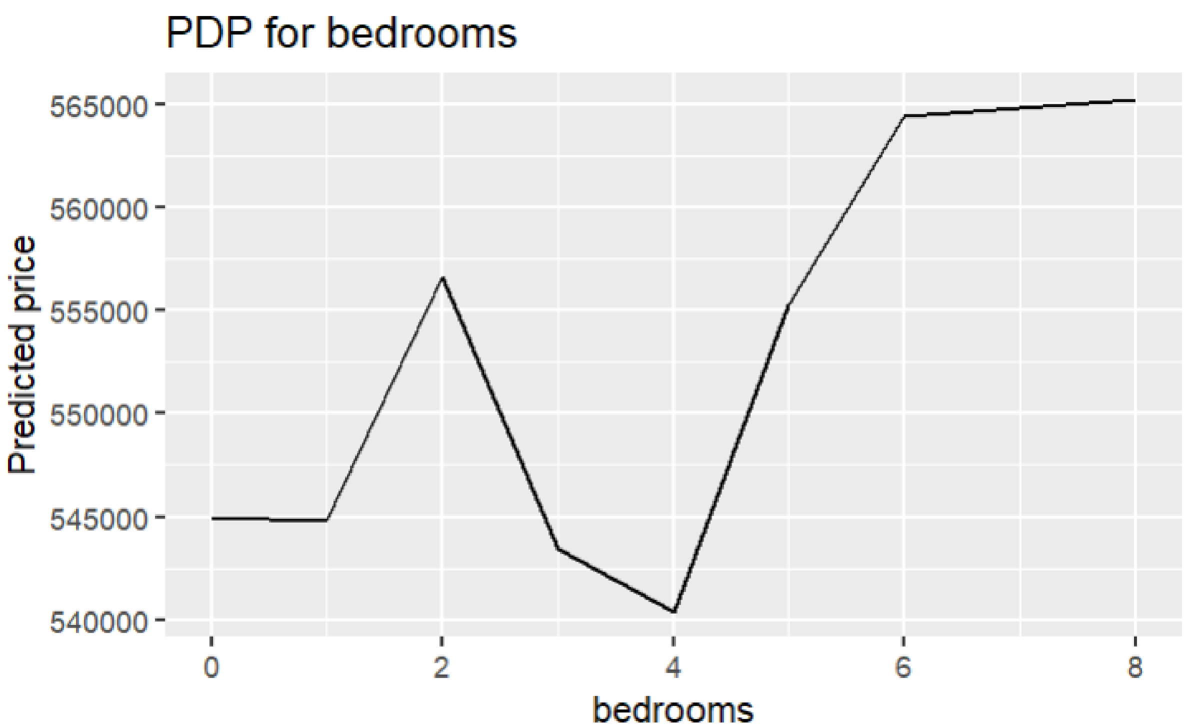
Metric	Value
Trees	100
Sample size	1,500 (random subsample)
Target variable	price (USD)
Features used	bedrooms, bathrooms, sqft_living, sqft_lot, floors, yr_built

PDP Results

Feature	Effect	Key finding
sqft_living	Strong positive	Dominant predictor; curve steepens sharply above ~6,500 sq ft
bathrooms	Positive	Gradual rise up to 5 bathrooms; sharp jump at 5.5+ (luxury homes)
bedrooms	Weak / noisy	Limited marginal effect; correlated with sqft_living
floors	Moderate positive	Increases up to ~3 floors, then stabilises or slightly decreases

Plots



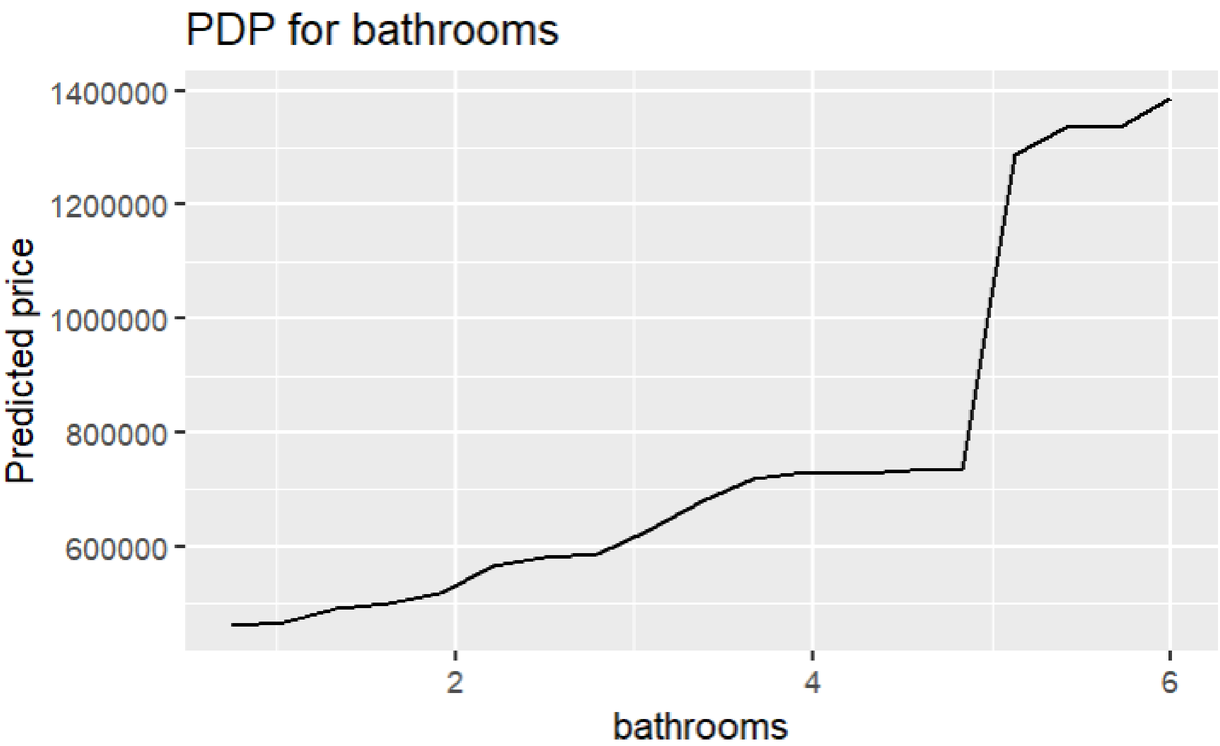


The PDP for bedrooms shows a much weaker relationship with price compared to sqft_living. The predicted price does not increase smoothly as the number of bedrooms increases.

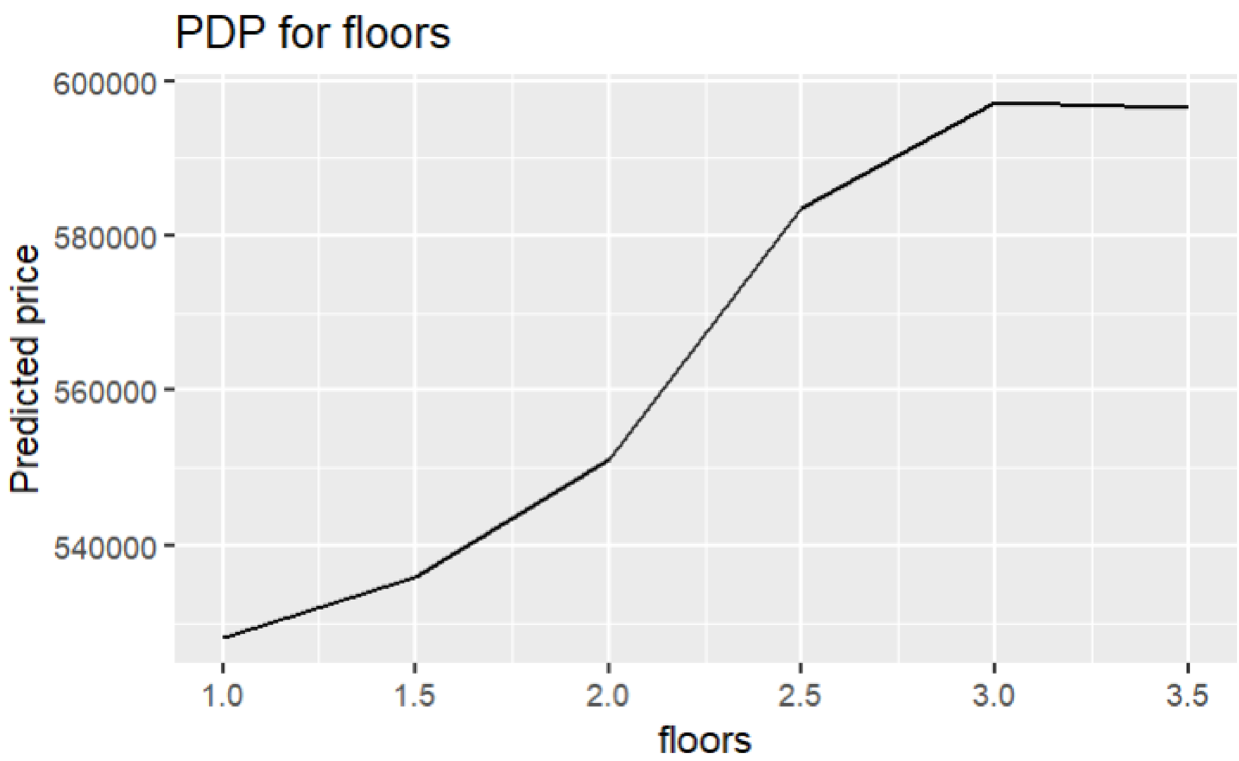
From 0 to around 4 bedrooms, the curve fluctuates considerably, with both increases and decreases in the predicted price. These irregularities suggest that the number of bedrooms alone is not a highly informative variable for the model. This behaviour may also be caused by correlations with other variables such as sqft_living and bathrooms.

After approximately 5 bedrooms, the predicted price increases and then stabilizes around 6–8 bedrooms. This suggests that houses with a larger number of bedrooms tend to have slightly higher prices, although the effect is relatively moderate.

The instability of the curve may also indicate that some bedroom configurations are uncommon in the dataset, producing noisier predictions. Overall, the graph suggests that bedrooms has a limited influence on price when compared with other variables.



The PDP for bathrooms shows a strong positive relationship with house prices. As the number of bathrooms increases, the predicted price generally increases as well.
The increase is relatively gradual between 1 and 5 bathrooms. However, around 5.5 bathrooms, the curve rises very sharply, indicating that luxury houses with many bathrooms are associated with much higher predicted prices.
After this large increase, the curve becomes more stable again, suggesting that additional bathrooms beyond this point still contribute positively but with smaller marginal gains.
This graph indicates that the model considers the number of bathrooms an important predictor of house value. The sharp jump may also reflect the presence of expensive high-end properties in the dataset, where a large number of bathrooms is associated with premium houses.



The PDP for floors shows a moderate positive relationship between the number of floors and the predicted price.
The predicted price increases steadily from 1 floor to around 3 floors, suggesting that multi-story houses tend to be more expensive than single-story houses. The largest increase appears between 2 and 2.5 floors.
After approximately 3 floors, the curve stabilizes and slightly decreases. This may indicate that adding more floors beyond a certain point does not substantially increase the predicted value of the property.
Compared with sqft_living and bathrooms, the influence of floors is smaller and more stable. Therefore, the number of floors appears to have a moderate but not dominant effect on house prices.